

Computer Architecture- Quiz #4-C

Name: Fatima Sohail

Roll#: 1230633

Section: 0

Total Time: 20 minutes

Total Marks: 20

7th April, 2026

CLO 2:

Question 1

Marks 10

Consider the following RISC-V code. For each instruction, translate it into machine language and represent the result in **binary**, **hexadecimal** formats. You are also required to clearly identify and label all fields of machine instruction such as **opcode**, **rd**, **rs1**, **rs2**, **funct3**, **funct7**, and **immediate** required for machine tra

Address	Instructions																					
0x00000004	L1: sub t0,t1,t2 <table><tr><td></td><td>Funct7</td><td>RS2</td><td>RS1</td><td>Funct3</td><td>Rd1</td><td>opcode</td></tr><tr><td>Binary</td><td>0000000</td><td>0000000</td><td>0000000</td><td>000</td><td>0000000</td><td>0110011</td></tr><tr><td>Hex</td><td colspan="6">00000003 006382B3</td></tr></table>		Funct7	RS2	RS1	Funct3	Rd1	opcode	Binary	0000000	0000000	0000000	000	0000000	0110011	Hex	00000003 006382B3					
	Funct7	RS2	RS1	Funct3	Rd1	opcode																
Binary	0000000	0000000	0000000	000	0000000	0110011																
Hex	00000003 006382B3																					
0x00000008	addi s2,s1,12 <table><tr><td></td><td colspan="5">rs1 3 rd</td><td>opcode</td></tr><tr><td>Binary</td><td>000000001100</td><td>00001000</td><td>00001000</td><td>0010011</td><td></td><td></td></tr><tr><td>Hex</td><td colspan="6">00000003 00C08113</td></tr></table>		rs1 3 rd					opcode	Binary	000000001100	00001000	00001000	0010011			Hex	00000003 00C08113					
	rs1 3 rd					opcode																
Binary	000000001100	00001000	00001000	0010011																		
Hex	00000003 00C08113																					
0x0000000C	lw t2,8(s2) <table><tr><td></td><td colspan="5">rs1 fun3 rd</td><td>opcode</td></tr><tr><td>Binary</td><td>000000001000</td><td>10010</td><td>010</td><td>000011</td><td></td><td></td></tr><tr><td>Hex</td><td colspan="6">00892323</td></tr></table>		rs1 fun3 rd					opcode	Binary	000000001000	10010	010	000011			Hex	00892323					
	rs1 fun3 rd					opcode																
Binary	000000001000	10010	010	000011																		
Hex	00892323																					
0x00000010	sw t2,-6(s2) <table><tr><td></td><td>imm.</td><td>rs2</td><td>rs1</td><td>fun3</td><td>imm4to.</td><td>opcode</td></tr><tr><td>Binary</td><td>11111111 00011</td><td>10010</td><td>010</td><td>1010</td><td>010011</td><td></td></tr><tr><td>Hex</td><td colspan="6">7F 7F3C9523</td></tr></table>		imm.	rs2	rs1	fun3	imm4to.	opcode	Binary	11111111 00011	10010	010	1010	010011		Hex	7F 7F3C9523					
	imm.	rs2	rs1	fun3	imm4to.	opcode																
Binary	11111111 00011	10010	010	1010	010011																	
Hex	7F 7F3C9523																					
0x00000014	beq s0,t5,L1 = 02 <table><tr><td></td><td>imm12,0:5</td><td>rs2</td><td>rs1</td><td>fun3</td><td>imm4:1,11</td><td>opcode</td></tr><tr><td>Binary</td><td>00000000 11110</td><td>01000</td><td>000</td><td>10100</td><td>1100011</td><td></td></tr><tr><td>Hex</td><td colspan="6">01E40A63</td></tr></table>		imm12,0:5	rs2	rs1	fun3	imm4:1,11	opcode	Binary	00000000 11110	01000	000	10100	1100011		Hex	01E40A63					
	imm12,0:5	rs2	rs1	fun3	imm4:1,11	opcode																
Binary	00000000 11110	01000	000	10100	1100011																	
Hex	01E40A63																					

14_H = ~~00000001~~

0000 0001 0100

ED

~~11111111 1010~~

CLO 2:

Question 2

Marks 10

Consider the following machine code and convert it to equivalent RISC-V code																						
Address	Instruction																					
0x00000004	0x0124E433 <table><tr><th></th><th>7</th><th>rs2</th><th>rs1</th><th>fun3</th><th>rd</th><th>opcode</th></tr><tr><td>Binary</td><td>00000001</td><td>0010</td><td>0100</td><td>110</td><td>0100</td><td>0011</td></tr><tr><td>Decimal</td><td>0</td><td>1</td><td>8</td><td>9</td><td>6</td><td>51</td></tr></table> Final Instruction: $or\ x8, x18, x9$		7	rs2	rs1	fun3	rd	opcode	Binary	00000001	0010	0100	110	0100	0011	Decimal	0	1	8	9	6	51
	7	rs2	rs1	fun3	rd	opcode																
Binary	00000001	0010	0100	110	0100	0011																
Decimal	0	1	8	9	6	51																
0x00000008	0xFF430293 <table><tr><th></th><th>fun3</th><th>rd</th><th>opcode</th></tr><tr><td>Binary</td><td>11111110</td><td>01000011</td><td>00000010</td></tr><tr><td>Decimal</td><td>-12</td><td>6</td><td>5</td></tr></table> Final Instruction: $addi\ x5, x6, -12$		fun3	rd	opcode	Binary	11111110	01000011	00000010	Decimal	-12	6	5									
	fun3	rd	opcode																			
Binary	11111110	01000011	00000010																			
Decimal	-12	6	5																			
0x0000000C	0xFF33AD23 <table><tr><th></th><th>rs2</th><th>rs1</th><th>3</th><th>imm4</th><th>opcode</th></tr><tr><td>Binary</td><td>11111110</td><td>00110011</td><td>1010</td><td>1010</td><td>0011</td></tr><tr><td>Decimal</td><td>6</td><td>14</td><td>5</td><td>1</td><td>35</td></tr></table> Final Instruction: $sw\ x6, -6(x14)$		rs2	rs1	3	imm4	opcode	Binary	11111110	00110011	1010	1010	0011	Decimal	6	14	5	1	35			
	rs2	rs1	3	imm4	opcode																	
Binary	11111110	00110011	1010	1010	0011																	
Decimal	6	14	5	1	35																	
0x00000010	0x01042E03 <table><tr><th></th><th>rs2</th><th>rs1</th><th>fun3</th><th>rd</th><th>opcode</th></tr><tr><td>Binary</td><td>00000001</td><td>00000100</td><td>0101</td><td>110</td><td>0000</td></tr><tr><td>Decimal</td><td>1</td><td>8</td><td>2</td><td>28</td><td>3</td></tr></table> Final Instruction: $lw\ x8, 0(x8)$		rs2	rs1	fun3	rd	opcode	Binary	00000001	00000100	0101	110	0000	Decimal	1	8	2	28	3			
	rs2	rs1	fun3	rd	opcode																	
Binary	00000001	00000100	0101	110	0000																	
Decimal	1	8	2	28	3																	
0x00000014	0xFE8F18E3 <table><tr><th></th><th>rs2</th><th>rs1</th><th>3</th><th>5</th><th>opcode</th></tr><tr><td>Binary</td><td>11111110</td><td>10001111</td><td>0001</td><td>1000</td><td>1100</td></tr><tr><td>Decimal</td><td>-16</td><td>28</td><td>30</td><td>1</td><td>99</td></tr></table> Final Instruction: $bne\ x30, x8, (-16)$		rs2	rs1	3	5	opcode	Binary	11111110	10001111	0001	1000	1100	Decimal	-16	28	30	1	99			
	rs2	rs1	3	5	opcode																	
Binary	11111110	10001111	0001	1000	1100																	
Decimal	-16	28	30	1	99																	

some label.

Computer Architecture- Quiz #4-C

Name: Aabia Ali

Roll#: 23r-0704

Section: D

Total Time: 20 minutes

Total Marks: 20

7th April, 2026

CLO 2:

Question 1

Marks 10

Consider the following RISC-V code. For each instruction, translate it into machine language and represent the result in **binary**, **hexadecimal** formats. You are also required to clearly identify and label all fields of machine instruction such as **opcode**, **rd**, **rs1**, **rs2**, **funct3**, **funct7**, and **immediate** required for machine tra

Address	Instructions																					
0x00000004	L1: sub t0, t1, t2 _{rd rs1 rs2} <table><tr><th></th><th>Funct7</th><th>RS2</th><th>RS1</th><th>Funct3</th><th>Rd1</th><th>opcode</th></tr><tr><td>Binary</td><td>01000000</td><td>00111</td><td>00110</td><td>000</td><td>00101</td><td>0110011</td></tr><tr><td>Hex</td><td>0x407302B3</td><td></td><td></td><td></td><td></td><td></td></tr></table>		Funct7	RS2	RS1	Funct3	Rd1	opcode	Binary	01000000	00111	00110	000	00101	0110011	Hex	0x407302B3					
	Funct7	RS2	RS1	Funct3	Rd1	opcode																
Binary	01000000	00111	00110	000	00101	0110011																
Hex	0x407302B3																					
0x00000008	addi s2, s1, 12 _{rd rs1 imm4:0} <table><tr><th></th><th>Funct7</th><th>RS2</th><th>RS1</th><th>Funct3</th><th>Rd1</th><th>opcode</th></tr><tr><td>Binary</td><td>00000000</td><td>00100</td><td>01001</td><td>000</td><td>10010</td><td>0110011</td></tr><tr><td>Hex</td><td>0x00C48933</td><td></td><td></td><td></td><td></td><td></td></tr></table>		Funct7	RS2	RS1	Funct3	Rd1	opcode	Binary	00000000	00100	01001	000	10010	0110011	Hex	0x00C48933					
	Funct7	RS2	RS1	Funct3	Rd1	opcode																
Binary	00000000	00100	01001	000	10010	0110011																
Hex	0x00C48933																					
0x0000000C	lw t2, 8(s2) _{rd rs1 imm4:0} <table><tr><th></th><th>Funct7</th><th>RS2</th><th>RS1</th><th>Funct3</th><th>Rd1</th><th>opcode</th></tr><tr><td>Binary</td><td>00000000</td><td>00100</td><td>10010</td><td>010</td><td>00111</td><td>0000011</td></tr><tr><td>Hex</td><td>0x00892383</td><td></td><td></td><td></td><td></td><td></td></tr></table>		Funct7	RS2	RS1	Funct3	Rd1	opcode	Binary	00000000	00100	10010	010	00111	0000011	Hex	0x00892383					
	Funct7	RS2	RS1	Funct3	Rd1	opcode																
Binary	00000000	00100	10010	010	00111	0000011																
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0x00000010	sw t2, -6(s2) _{rs1 rs2 imm4:0} <table><tr><th></th><th>Funct7</th><th>RS2</th><th>RS1</th><th>Funct3</th><th>Rd1</th><th>opcode</th></tr><tr><td>Binary</td><td>11111110</td><td>00111</td><td>010</td><td>000</td><td>10010</td><td>0100011</td></tr><tr><td>Hex</td><td>0xF43A923</td><td></td><td></td><td></td><td></td><td></td></tr></table>		Funct7	RS2	RS1	Funct3	Rd1	opcode	Binary	11111110	00111	010	000	10010	0100011	Hex	0xF43A923					
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Binary	11111110	00111	010	000	10010	0100011																
Hex	0xF43A923																					
0x00000014	beq s0, t5, L1 _{rs1 rs2} <table><tr><th></th><th>Funct7</th><th>RS2</th><th>RS1</th><th>Funct3</th><th>Rd1</th><th>opcode</th></tr><tr><td>Binary</td><td>0000000</td><td>11110</td><td>01000</td><td>000</td><td>10000</td><td>1100011</td></tr><tr><td>Hex</td><td>0x0308A9</td><td></td><td></td><td></td><td></td><td></td></tr></table> 0x00000014 0x01E40863		Funct7	RS2	RS1	Funct3	Rd1	opcode	Binary	0000000	11110	01000	000	10000	1100011	Hex	0x0308A9					
	Funct7	RS2	RS1	Funct3	Rd1	opcode																
Binary	0000000	11110	01000	000	10000	1100011																
Hex	0x0308A9																					

imm = 14 - 4

= 14 - 4

= 14 - 4

= 0x10 = 00100000

12 11 10:6 4:1
00000000100000

Page 1 of 2

imm = 14

= 4 - 14

= -10

= ffffffff0

-6 = 1111111010

imm 0 rs2 rs1 f3 imm4:0 opcode
111111 0011 10010 010 11010 0100011

hex = 0xf2792D23

CLO 2:

Question 2

Marks 10

Consider the following machine code and convert it to equivalent RISC-V code

Address	Instruction																					
0x00000004	0x0124E433 <table><tr><th></th><th>f7</th><th>rs2</th><th>rs1</th><th>f3</th><th>rd</th><th>opcode</th></tr><tr><td>Binary</td><td>0000000</td><td>10010</td><td>01001</td><td>110</td><td>0100000</td><td>0110011</td></tr><tr><td>Decimal</td><td>0</td><td>18</td><td>9</td><td>6</td><td>8</td><td>51</td></tr></table> Final Instruction: or s0, s1, s2		f7	rs2	rs1	f3	rd	opcode	Binary	0000000	10010	01001	110	0100000	0110011	Decimal	0	18	9	6	8	51
	f7	rs2	rs1	f3	rd	opcode																
Binary	0000000	10010	01001	110	0100000	0110011																
Decimal	0	18	9	6	8	51																
0x00000008	0xFF430293 <table><tr><th></th><th>imm=11:0</th><th>rs1</th><th>f3</th><th>rd</th><th>opcode</th></tr><tr><td>Binary</td><td>11111110100</td><td>00110</td><td>000</td><td>00101</td><td>0010011</td></tr><tr><td>Decimal</td><td>-12</td><td>6</td><td>0</td><td>5</td><td>19</td></tr></table> Final Instruction: addi t0, t1, -12		imm=11:0	rs1	f3	rd	opcode	Binary	11111110100	00110	000	00101	0010011	Decimal	-12	6	0	5	19			
	imm=11:0	rs1	f3	rd	opcode																	
Binary	11111110100	00110	000	00101	0010011																	
Decimal	-12	6	0	5	19																	
0x0000000C	0xFF33AD23 <table><tr><th></th><th>imm=11:5</th><th>rs2</th><th>rs1</th><th>f3</th><th>imm=4:0</th><th>opcode</th></tr><tr><td>Binary</td><td>1111111</td><td>10011</td><td>00111</td><td>010</td><td>11010</td><td>0100011</td></tr><tr><td>Decimal</td><td>13</td><td>7</td><td>7</td><td>2</td><td>26</td><td>35</td></tr></table> Final Instruction: sw t2, -6(a3)		imm=11:5	rs2	rs1	f3	imm=4:0	opcode	Binary	1111111	10011	00111	010	11010	0100011	Decimal	13	7	7	2	26	35
	imm=11:5	rs2	rs1	f3	imm=4:0	opcode																
Binary	1111111	10011	00111	010	11010	0100011																
Decimal	13	7	7	2	26	35																
0x00000010	0x01042E03 <table><tr><th></th><th>imm=11:0</th><th>rs1</th><th>f3</th><th>rd</th><th>opcode</th></tr><tr><td>Binary</td><td>00000010000</td><td>01000</td><td>010</td><td>11100</td><td>0000011</td></tr><tr><td>Decimal</td><td>16</td><td>8</td><td>2</td><td>28</td><td>3</td></tr></table> Final Instruction: lw t3, 16(s0)		imm=11:0	rs1	f3	rd	opcode	Binary	00000010000	01000	010	11100	0000011	Decimal	16	8	2	28	3			
	imm=11:0	rs1	f3	rd	opcode																	
Binary	00000010000	01000	010	11100	0000011																	
Decimal	16	8	2	28	3																	
0x00000014	0xFF8F18E3 <table><tr><th></th><th>imm=10:5</th><th>rs2</th><th>rs1</th><th>f3</th><th>imm=4:1</th><th>opcode</th></tr><tr><td>Binary</td><td>1111111</td><td>01000</td><td>11110</td><td>000</td><td>10000</td><td>1010011</td></tr><tr><td>Decimal</td><td>8</td><td>8</td><td>30</td><td>1</td><td>16</td><td>33</td></tr></table> Final Instruction: bne t5, s0, 0b101111110000		imm=10:5	rs2	rs1	f3	imm=4:1	opcode	Binary	1111111	01000	11110	000	10000	1010011	Decimal	8	8	30	1	16	33
	imm=10:5	rs2	rs1	f3	imm=4:1	opcode																
Binary	1111111	01000	11110	000	10000	1010011																
Decimal	8	8	30	1	16	33																

Computer Architecture- Quiz #4-C

Name: ABDUL GHUFRAN Roll#: 23i-0576

Section: C

Total Time: 20 minutes

Total Marks: 20

7th April, 2026

CLO 2:

Question 1

Marks 10

Consider the following RISC-V code. For each instruction, translate it into machine language and represent the result in **binary**, **hexadecimal** formats. You are also required to clearly identify and label all fields of machine instruction such as **opcode**, **rd**, **rs1**, **rs2**, **funct3**, **funct7**, and **immediate** required for machine tra

and immediate required for machine tra							
Address	Instructions						
0x00000004	L1: sub t0,t1,t2						
R-Type		Funct7	RS2	RS1	Funct3	Rd1	opcode
	Binary	0100000	00111	00110	000	00101	0110011
	Hex	0x40730283					0010011
0x00000008	addi s2,s1,12						
I type		imm _{11:0}	rs1	Funct3	rd1	opcode	
	Binary	0000001100	01001	000	0010	0010011	
	Hex	00892383					
0x0000000C	lw t2,8(s2)						
I X-type		imm _{11:5}	rs2	rs1	Funct3	imm _{4:0}	opcode
	Binary	0000000	10010	00111	010	01000	0000011
	Hex						
0x00000010	sw t2,-6(s2)						
S-X-type S		imm _{11:5}	rs2	rs1	Funct3	imm _{4:0}	opcode
	Binary	1111111	10010	00111	010	11010	0100011
	Hex	FF13AD2B					
0x00000014	beq s0,t5,L1						
B-type		imm _{12:5}	rs2	rs1	Funct3	imm _{4:1}	opcode
	Binary	00000001110	01000	000	000	00100	1100011
	Hex	01E46263					

imm_{11:0} rs1 Funct3 rd1
 0000 00001000 0011 010 10010
 0123A403

CLO 2:

Question 2

Marks 10

Consider the following machine code and convert it to equivalent RISC-V code

Address	Instruction								
0x00000004	0x0124E433 <i>R-type</i> <table border="1"> <tr> <td></td><td>0000 0001 0010 0100 1110 0100 0011 0011</td></tr> <tr> <td></td><td>Func1 R2 R31 Funct3 R2 opcode</td></tr> <tr> <td>Binary</td><td>0000000100100100110010000110011</td></tr> <tr> <td>Decimal</td><td>0 18 9 6 8 51</td></tr> </table> Final Instruction:		0000 0001 0010 0100 1110 0100 0011 0011		Func1 R2 R31 Funct3 R2 opcode	Binary	0000000100100100110010000110011	Decimal	0 18 9 6 8 51
	0000 0001 0010 0100 1110 0100 0011 0011								
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Decimal	0 18 9 6 8 51								
0x00000008	0xFF430293 <i>I-type</i> <table border="1"> <tr> <td></td><td>1111 1111 0000 0011 0000 0010 1001 0011</td></tr> <tr> <td></td><td>imm16:0 R31 Funct3 R2 opcode</td></tr> <tr> <td>Binary</td><td>11111111010000110000001010010011</td></tr> <tr> <td>Decimal</td><td>6 0 5 19</td></tr> </table> Final Instruction:		1111 1111 0000 0011 0000 0010 1001 0011		imm16:0 R31 Funct3 R2 opcode	Binary	11111111010000110000001010010011	Decimal	6 0 5 19
	1111 1111 0000 0011 0000 0010 1001 0011								
	imm16:0 R31 Funct3 R2 opcode								
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Decimal	6 0 5 19								
0x0000000C	0xFF33AD23 <i>S-type</i> <table border="1"> <tr> <td></td><td>1111 1111 0011 0011 1010 1101 0010 0011</td></tr> <tr> <td></td><td>imm14:5 R32 R31 Funct3 imm14:0 opcode</td></tr> <tr> <td>Binary</td><td>1111111100110011010110100011</td></tr> <tr> <td>Decimal</td><td>35</td></tr> </table> Final Instruction:		1111 1111 0011 0011 1010 1101 0010 0011		imm14:5 R32 R31 Funct3 imm14:0 opcode	Binary	1111111100110011010110100011	Decimal	35
	1111 1111 0011 0011 1010 1101 0010 0011								
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0x00000010	0x01042E03 <i>S-type</i> <table border="1"> <tr> <td></td><td>0000 0001 0000 0100 0010 1110 0000 0011</td></tr> <tr> <td></td><td>imm14:5 R32 R31 Funct3 imm14:0 opcode</td></tr> <tr> <td>Binary</td><td>00000001000001000010111000000011</td></tr> <tr> <td>Decimal</td><td>3</td></tr> </table> Final Instruction:		0000 0001 0000 0100 0010 1110 0000 0011		imm14:5 R32 R31 Funct3 imm14:0 opcode	Binary	00000001000001000010111000000011	Decimal	3
	0000 0001 0000 0100 0010 1110 0000 0011								
	imm14:5 R32 R31 Funct3 imm14:0 opcode								
Binary	00000001000001000010111000000011								
Decimal	3								
0x00000014	0xFE8F18E3 <i>B-type</i> <table border="1"> <tr> <td></td><td>1111 1110 0100 1111 0001 0100 1110 0011</td></tr> <tr> <td></td><td>imm12,10:5 R32 R31 Funct3 imm14,11 opcode</td></tr> <tr> <td>Binary</td><td>11111110010011110001010011100011</td></tr> <tr> <td>Decimal</td><td>99</td></tr> </table> Final Instruction: <i>1110001</i>		1111 1110 0100 1111 0001 0100 1110 0011		imm12,10:5 R32 R31 Funct3 imm14,11 opcode	Binary	11111110010011110001010011100011	Decimal	99
	1111 1110 0100 1111 0001 0100 1110 0011								
	imm12,10:5 R32 R31 Funct3 imm14,11 opcode								
Binary	11111110010011110001010011100011								
Decimal	99								

Sw

Lw

bne

Computer Architecture- Quiz #4-C

Name: Abdul Rauf Roll#: 231-0591

Section: C

Total Time: 20 minutes Total Marks: 20 7th April, 2026

CLO 2:

Question 1

Marks 10

Consider the following RISC-V code. For each instruction, translate it into machine language and represent the result in **binary**, **hexadecimal** formats. You are also required to clearly identify and label all fields of machine instruction such as **opcode**, **rd**, **rs1**, **rs2**, **funct3**, **funct7**, and **immediate** required for machine tra

Address	Instructions																					
0x00000004	L1: sub t0,t1,t2 <table><tr><th></th><th>Funct7</th><th>RS2</th><th>RS1</th><th>Funct3</th><th>Rd1</th><th>opcode</th></tr><tr><td>Binary</td><td>0100000</td><td>00110</td><td>00110</td><td>000</td><td>00100</td><td>0110011</td></tr><tr><td>Hex</td><td></td><td></td><td></td><td></td><td></td><td>0x33</td></tr></table>		Funct7	RS2	RS1	Funct3	Rd1	opcode	Binary	0100000	00110	00110	000	00100	0110011	Hex						0x33
	Funct7	RS2	RS1	Funct3	Rd1	opcode																
Binary	0100000	00110	00110	000	00100	0110011																
Hex						0x33																
0x00000008	addi s2,s1,12 <table><tr><th></th><th>imm 11:0</th><th>rs1</th><th>funct3</th><th>rd</th><th>opcode</th></tr><tr><td>Binary</td><td>00000001100</td><td>0001</td><td>000</td><td>0010</td><td>0010011</td></tr><tr><td>Hex</td><td></td><td></td><td></td><td></td><td>0x13</td></tr></table>		imm 11:0	rs1	funct3	rd	opcode	Binary	00000001100	0001	000	0010	0010011	Hex					0x13			
	imm 11:0	rs1	funct3	rd	opcode																	
Binary	00000001100	0001	000	0010	0010011																	
Hex					0x13																	
0x0000000C	lw t2,8(s2) <table><tr><th></th><th>imm 11:0</th><th>rs1</th><th>funct3</th><th>rd</th><th>opcode</th></tr><tr><td>Binary</td><td>00000001000</td><td>0001</td><td>010</td><td>0010</td><td>0000011</td></tr><tr><td>Hex</td><td></td><td></td><td></td><td></td><td>0x10</td></tr></table>		imm 11:0	rs1	funct3	rd	opcode	Binary	00000001000	0001	010	0010	0000011	Hex					0x10			
	imm 11:0	rs1	funct3	rd	opcode																	
Binary	00000001000	0001	010	0010	0000011																	
Hex					0x10																	
0x00000010	sw t2,-6(s2) <table><tr><th></th><th>imm 11:5</th><th>rs2</th><th>rs1</th><th>funct3</th><th>imm 4:0</th><th>opcode</th></tr><tr><td>Binary</td><td>111111</td><td>00110</td><td>010</td><td>1110</td><td>0100011</td><td></td></tr><tr><td>Hex</td><td></td><td></td><td></td><td></td><td></td><td>0x23</td></tr></table>		imm 11:5	rs2	rs1	funct3	imm 4:0	opcode	Binary	111111	00110	010	1110	0100011		Hex						0x23
	imm 11:5	rs2	rs1	funct3	imm 4:0	opcode																
Binary	111111	00110	010	1110	0100011																	
Hex						0x23																
0x00000014	beq s0,t5,L1 <table><tr><th></th><th>imm 12,10:5</th><th>rs2</th><th>rs1</th><th>funct3</th><th>imm 4:1</th><th>opcode</th></tr><tr><td>Binary</td><td>111111</td><td>11110</td><td>0000</td><td>000</td><td>1000</td><td>1100011</td></tr><tr><td>Hex</td><td></td><td></td><td></td><td></td><td></td><td>0x63</td></tr></table>		imm 12,10:5	rs2	rs1	funct3	imm 4:1	opcode	Binary	111111	11110	0000	000	1000	1100011	Hex						0x63
	imm 12,10:5	rs2	rs1	funct3	imm 4:1	opcode																
Binary	111111	11110	0000	000	1000	1100011																
Hex						0x63																

CLO 2:

Question 2

Marks 10

Consider the following machine code and convert it to equivalent RISC-V code																						
Address	Instruction																					
0x00000004	0x0124E433																					
R-Type	<table><tr><td></td><td>func7</td><td colspan="5">opcode</td></tr><tr><td>Binary</td><td>0000 000</td><td>0010 0100</td><td>1110 0100</td><td>0011 0011</td><td>0011</td><td></td></tr><tr><td>Decimal</td><td></td><td>func7</td><td>rs2</td><td>rs1</td><td>func3</td><td>rd</td></tr></table> OR x0, x1, x2 Final Instruction: OR x8, x9, x18.		func7	opcode					Binary	0000 000	0010 0100	1110 0100	0011 0011	0011		Decimal		func7	rs2	rs1	func3	rd
	func7	opcode																				
Binary	0000 000	0010 0100	1110 0100	0011 0011	0011																	
Decimal		func7	rs2	rs1	func3	rd																
0x00000008	0xFF430293																					
I-Type	<table><tr><td></td><td>imm11</td><td>rs1</td><td>func3</td><td>rd</td><td>opcode</td></tr><tr><td>Binary</td><td>1111 1111 0100</td><td>0011 0000</td><td>0010 1001</td><td>0011</td><td></td></tr><tr><td>Decimal</td><td></td><td></td><td></td><td></td><td></td></tr></table> addi t0, t1, -12 Final Instruction: addi x5, x6, 0xFF4.		imm11	rs1	func3	rd	opcode	Binary	1111 1111 0100	0011 0000	0010 1001	0011		Decimal								
	imm11	rs1	func3	rd	opcode																	
Binary	1111 1111 0100	0011 0000	0010 1001	0011																		
Decimal																						
0x0000000C	0xFF33AD23																					
S-Type	<table><tr><td></td><td>imm11</td><td>rs2</td><td>rs1</td><td>imm4</td><td>opcode</td></tr><tr><td>Binary</td><td>1111 1111 0011</td><td>0011 1010</td><td>1101 0010</td><td>0011</td><td></td></tr><tr><td>Decimal</td><td></td><td></td><td></td><td></td><td></td></tr></table> sw t2, -6(S3) Final Instruction: sw x7, 0xFFA(x19)		imm11	rs2	rs1	imm4	opcode	Binary	1111 1111 0011	0011 1010	1101 0010	0011		Decimal								
	imm11	rs2	rs1	imm4	opcode																	
Binary	1111 1111 0011	0011 1010	1101 0010	0011																		
Decimal																						
0x00000010	0x01042E03																					
I-Type	<table><tr><td></td><td>imm11</td><td>rs2</td><td>rs1</td><td>func3</td><td>rd</td><td>opcode</td></tr><tr><td>Binary</td><td>0000 0001</td><td>0000 0100</td><td>0010 1110</td><td>0000</td><td>0011</td><td></td></tr><tr><td>Decimal</td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table> lw t3, 16(S0) Final Instruction: lw x28, 16(x8)		imm11	rs2	rs1	func3	rd	opcode	Binary	0000 0001	0000 0100	0010 1110	0000	0011		Decimal						
	imm11	rs2	rs1	func3	rd	opcode																
Binary	0000 0001	0000 0100	0010 1110	0000	0011																	
Decimal																						
0x00000014	0xFE8F18E3																					
	<table><tr><td></td><td>imm11</td><td>rs2</td><td>rs1</td><td>func3</td><td>imm4</td><td>opcode</td></tr><tr><td>Binary</td><td>1111 1111 1000</td><td>1111 0001</td><td>1000 1110</td><td>0011</td><td>0011</td><td></td></tr><tr><td>Decimal</td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table> bne. x80, x39, 0xFF8. Final Instruction: bne. x80, x39, 0xFF8.		imm11	rs2	rs1	func3	imm4	opcode	Binary	1111 1111 1000	1111 0001	1000 1110	0011	0011		Decimal						
	imm11	rs2	rs1	func3	imm4	opcode																
Binary	1111 1111 1000	1111 0001	1000 1110	0011	0011																	
Decimal																						

Ans:- $bne\ S0, t5, -16.$